National Institute of Technology, Rourkela

Name of the Examination: B. Tech. End-Semester (Spring 2021-22)

Branch	**: CS**	**Semester**	**: VI**
Title of the Course	**: Computer Networks**	**Course Code**	**: CS3002**
Time: 3 Hours		**Maximum Marks: 50**	

Instructions: 1. Answer **all** short-answer type questions from **Section-A**.

2. Answer any **four (4)** questions from **Section-B**.

3. All parts of a question must be answered at one place.

Section-A

(Attempt all short-answer type questions from this section)

1. Answer the following parts. **(10 x 1 = 10 Marks)**

(a) What is the optimality principle used for routing algorithm?

(b) Answer True or False for the following statements

(i) DHCP always assigns a client an IP address for an *indefinite* period of time.

(ii) The logical address will change from hop to hop, but the physical addresses usually remain the same?

(c) A class B network on the Internet has subnet mask of 255.255.240.0. What is the maximum number of hosts per subnet?

(d) What is an IP *Prefix?*

(e) What is loose source routing?

(f) Give any one advantage and any one disadvantage of the Flooding?

(g) Why *Traffic Shaping* is required in the network?

(h) What is *multicasting*?

(i) Explain the situation when the routing protocol leads to count-to-infinity problem?

(j) Give two reasons why Network Address Translation is so important?

Section-B

(Attempt any four from this section) **(10 x 4 =40 Marks)**

2. Answer the following.

(a) Derive an equation for channel efficiency of classic Ethernet in terms of the frame length, **F**, network bandwidth, **B**, the cable length, **L**, and the speed of signal propagation, **c**, for the optimal case of **e** contention slots per frame under conditions of heavy and constant load, that is, with **k** stations always ready to transmit. Assume a constant retransmission probability in each slot. **(5 Marks)**

(b) Explain Switched Ethernet using a neat diagram. Explain why CSMA/CD algorithm is not needed in it? **(5 Marks)**

3. Answer the following:

(a) Explain the following techniques: **(5 Marks)**

(i) Leaky bucket algorithm

(ii) Packet scheduling algorithm

(b) Explain Load Shedding approach used for network layer congestion control? **(5 Marks)**

4. Answer the following:

(a) Explain the following two TCP congestion control algorithms: **(5 Marks)**

(i) Slow Start

(ii) Fast Retransmit and Fast Recovery

(b) Explain Internet Control Message Protocol. Describe the important ICMP message types? **(5 Marks)**

5. Answer the following:

(a) Explain two-army problem. Explain the handshaking mechanism of TCP to release a connection between two hosts in a network? **(5 Marks)**

(b) Describe each of the fields of the TCP segment header format of Internet Transport Protocol in detail. **(5 Marks)**

6. Answer the following:

(a) Explain Hierarchical Routing using any suitable network example. Give advantages and disadvantages of this routing scheme? **(7 Marks)**

(b) We use CIDR. Without using longest prefix matching, a forwarding table looks like this. If we use longest prefix matching, we can combine a few entries together. What is a table with a minimum number of entries that still be able to forward packet correctly? **(3 Marks)**

Prefix	Outgoing Interface
128.0.0.0/11	eth1
128.16.0.0/12	eth1
128.24.0.0/12	eth2
128.32.0.0/12	eth2
128.40.0.0/12	eth1
128.48.0.0/11	eth1
128.64.0.0/9	eth0
128.128.0.0/10	eth0
128.160.0.0/11	eth1
128.176.0.0/11	eth0
128.192.0.0/9	eth0
default	eth3